CMP at the Wafer Edge – Modeling the Interaction between Wafer Edge Geometry and Polish Performance

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ABSTRACT

As the drive to improve integrated circuit manufacturing yield continues, renewed attention is being paid to the edge of the wafer. The industry is seeking to reduce the edge exclusion region and achieve good performance to within 2 mm or smaller. This creates substantial challenges, both for CMP and for the starting wafer. In this work, we consider two key elements that play a role in non-uniform polish near the edge of the wafer.

First, we study the impact of localized pressure on the edge of the wafer as a function of the wafer and retaining ring pressures, gap separation between wafer and retaining ring, and pad material properties (pad Young's modulus). Simulations show that several millimeters into the wafer from the edge can polish either more quickly or more slowly than the center of the wafer, depending on the combination of these parameters. Second, we also consider the impact of wafer edge roll-off (the specific thickness or front surface elevation of the wafer geometry) on polishing uniformity. We again find that the polish uniformity can be affected dramatically, depending on the details of the starting wafer geometry.

We believe that several innovations and optimizations are likely to arise in order to meet future wafer edge polish uniformity requirements. These include tool geometry and process improvements, tailoring of the pad material properties, and starting wafer edge geometry optimization and control.

INTRODUCTION

CMP is an enabling technology to achieve local planarization and wafer-level uniformity. Recent studies have shown the existence of an important topographic issue: wafer edge roll-off. The edge roll-off is typically a convex profile near the wafer periphery and occurs within 1-5 mm from the wafer edge, as illustrated in figure 1. These profiles can exhibit a range of variations other than a simple convex profile, as discussed later. In both prime wafer preparation and in IC fabrication, edge roll-off may result from CMP polish nonuniformity. In addition, this edge roll-off can potentially impact uniformity in subsequent CMP processes [1].

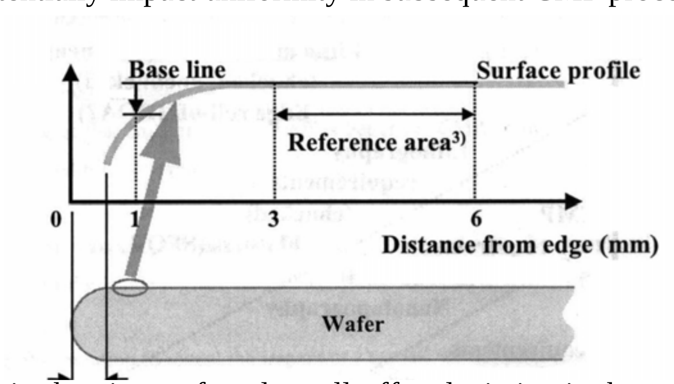

Figure 1. A schematic showing wafer edge roll-off, a deviation in the wafer geometry from a flat baseline level, near the edge of the wafer [1].

The edge roll-off can result from the inherent discontinuities of the process tool geometry at the wafer edge. Figure 2 shows the configuration of a typical rotary CMP tool, in which the wafer carrier holds the wafer facing down, which is polished against the polishing pad. Figure 2 also schematically pictures the geometry near the wafer edge. The wafer is held by a retaining ring, which is usually a few millimeters away. In a typical setting, different pressures are applied to the wafer and ring with the ring usually under higher pressure to prevent the wafer from slipping out. The pad bends around the wafer edge due to the existence of the gap and retaining ring; thus the wafer edge is polished non-uniformly due to a localized pressure concentration.

The factors causing the non-uniform pressure distribution are gap size, pad stiffness (effective Young's modulus), and pressures on the wafer and retaining ring. In this paper, we study how these factors influence the edge roll-off profile. First, we present some background and examples of wafer edge roll-off. Second, we review the contact wear model used in the simulations. Then, we present our simulation results in a three part discussion. The first part focuses on static cases, which is the relationship between the tool and pad factors and pressure distribution at a single time instant. Second we consider dynamic cases, where the evolution of edge profile is simulated over time with different values of the factors. The first two parts help us to understand how edge roll-off is generated; the third part examines the influence of initial wafer profile on subsequent polishing

Several studies have focused on the relationship between wafer scale CMP nonuniformity and tool geometry [1][2][3]. Fukuda et al. [1] found that wafer flatness in the peripheral area of a wafer was relatively worse than in the interior region, which strongly suggests that edge roll-off deteriorates wafer flatness at the wafer edge. They also found that an optimum polishing condition could be used with a range of wafers having limited roll-off variation. Castillo-Mejia et al. [2] developed a stress-based model for the qualitative description of removal rates during silicon dioxide CMP on Applied Materials' Mirra polisher, and studied the wafer-level CMP nonuniformity affected by the wafer-retaining ring separation and various pressures exerted by the retaining ring. Sorooshian et al. [3] demonstrated via experiments that variations in wafer geometry as measured by overall shape and size of wafer-ring gap can significantly impact the extent of within wafer non-uniformity in pressure and hence inter-layer dielectric removal rate.

Measurements of several wafer edge profiles are shown here to give us some intuition about the real wafer edge profile. The measurements are done on blanket starting wafers from various wafer vendors. The starting wafer is usually single-sided or double-sided polished after it is sliced from the bulk and ground. Thus, the measurements on starting wafers can exhibit edge roll-off profiles, and these initial topographic profiles will influence further CMP process.

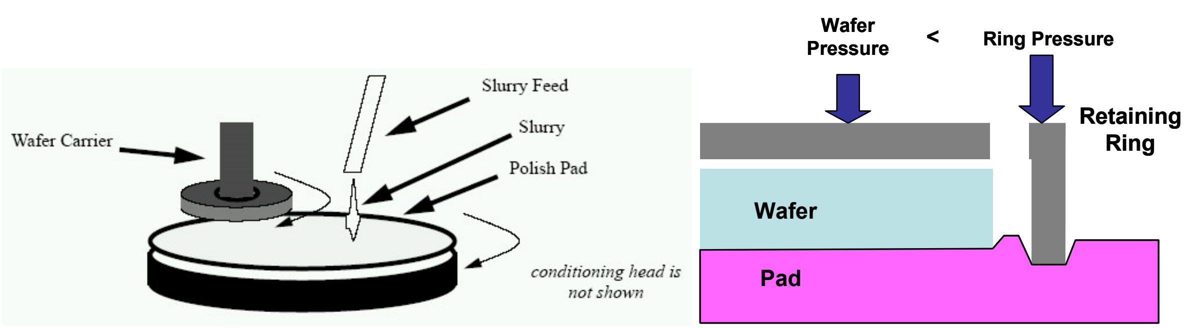

Figure 2. Illustrations of a typical rotary CMP tool configuration. The left diagram shows the whole tool, and the right diagram focuses on the geometry near the wafer edge.

Both the wafer front surface profile and the thickness are measured by ADE WaferSight optical flatness gauge. A dataset of 59 front surface scans and 100 thickness scans was examined, and a few typical profiles are shown in figure 3. The wafer front surface is the surface being polished and its profile is measured in unchucked condition. The profile shows mainly wafer warp and bow, and the edge of the wafer does not have any distinctive roll-off profile. In contrast, the thickness scans have much smaller magnitude but exhibit distinctive roll-off profiles near the edge of wafer. Because the wafer is under substantial pressure during the CMP process as schematically indicated in figure 4, we conjecture that the in-process wafer geometry variation at the pad/wafer interface is determined by the wafer thickness variation. The thickness profiles show many different patterns, which is probably due to the variety of CMP tools, polishing pads, and processing conditions used by different wafer vendors.

WAFER SCALE CONTACT WEAR MODEL

In this work, we apply a contact wear model to simulate CMP on the wafer scale. Contact wear modeling of CMP was proposed by Chekina [4], Yoshida [5], and Vlassak [6]. Its concept is to relate local pressures on the wafer surface to the polishing pad displacement, and use the local pressures to derive local removal rates assuming the polish rate is linearly proportional to the pressure. As the film surface evolves through time, the pad displacement is changed, and thus local pressures and removal rates are modified. The simulation surface is discretized into elements, and a time-stepped algorithm is used to determine the final post-CMP film surface. Yoshida [5] describes a boundary element method to solve the matrix form of this equation.

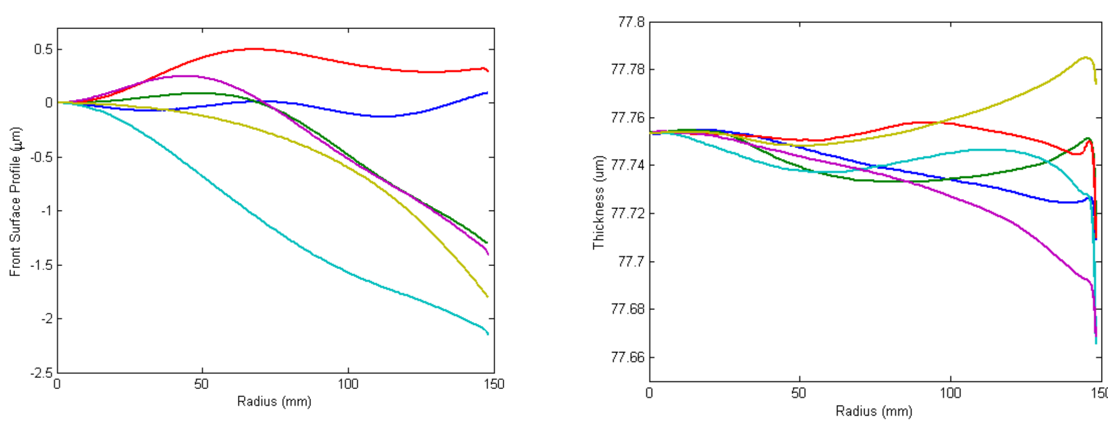

Figure 3. Left plot shows several measured front surface profiles; note that the range of variation is about a few microns. Right plot shows several measured thickness scans; here, the range of variation is approximately only 0.05 microns.

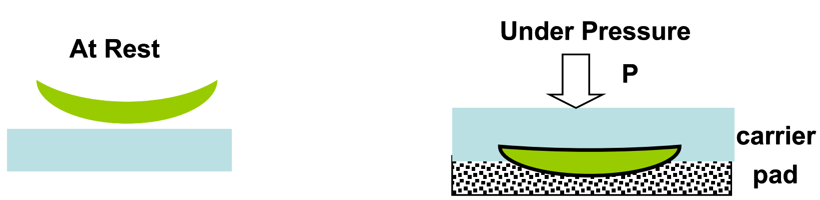

Figure 4. The diagrams show the difference between the wafer at rest and under pressure. We assume that the profile the pad sees is determined by the wafer thickness.

Applying 2D contact wear simulation to the whole wafer is computationally expensive, as we need to discretize a 300 mm wafer into sub-millimeter cells to study in detail the several millimeter area near the edge of the wafer. Instead we apply the model to an area of 60 mm x 60 mm at the wafer edge, as illustrated in figure 5. The area is discretized into 0.1 mm x 0.1 mm cells, and we study the 10 mm line across the gap at the center of the area. The distance from the area of interest to the closest boundary is 20 mm, and we assume that the interaction of elements more than 20 mm apart is neglectable. The assumption is reasonable as the interaction distance, of which planarization length can be one measure, is generally less than 10 mm, and in our simulation the pressure distribution and wear profile are stable when the area of area of interest is 10 mm inside the boundary.

One complication of the simulation is the different pressure on the retaining ring. The contact wear model is not able to capture the case directly, so an additional iterative loop is used to modify the displacements to achieve the two different pressures on the wafer and the ring.

SIMULATION AND RESULTS

To develop intuition as to how each factor influences the polishing at the edge of the wafer, we first study a set of static cases, in which we examine the pressure distribution response to different factors at one time instant. In figures 6-8, the same format is used: the gap starts at x=0, the wafer is on the left of the gap and the ring on the right, and the wafer surface is set to a vertical level of zero. The wafer is assumed to be flat, and drops to negative infinity at x=0.

First, we vary the effective Young's modulus, and keep the gap size fixed at 1 mm, wafer pressure at 4 psi, and ring pressure at 5 psi. We simulate for three different effective Young's modulus, 2 MPa, 20 MPa, and 40 MPa. The pressure distribution and pad profile is shown in figure 6. The pressure distributions we see that are the same for all values of Young's modulus, while the pad profiles are different; the softer the pad, the more the pad bends. The effective Young's modulus works as a scaling factor in the pad bending profile, without affecting the pressure distribution. Although in the static case the Young's modulus does not affect the pressure distribution, it does matter in the dynamic case, which will be discussed later.

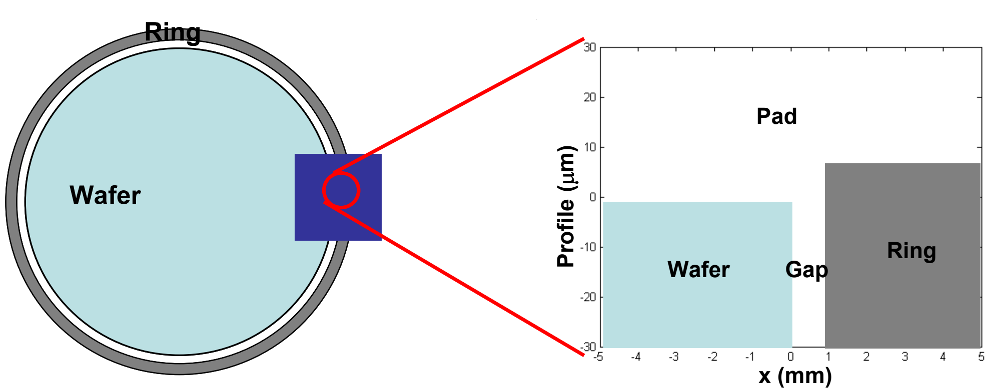

Figure 5. Illustration of simulated area, a 60 mm x 60 mm square centered at the wafer edge. We focus on the 10 mm line across the gap, as shown on the right.

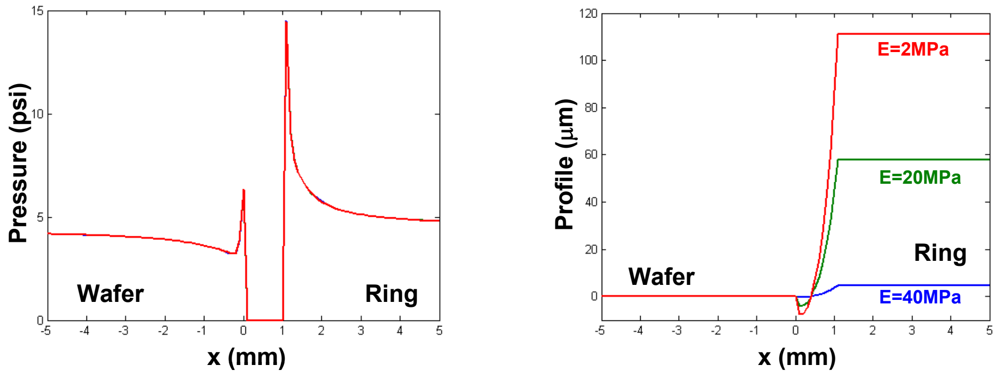

Figure 6. Static case to study the effect of Young's modulus. For all three simulations, gap is 1 mm, wafer pressure is 4 psi, and ring pressure is 5 psi.

Next, we study the relationship between the pressure distribution and gap size. We simulate for gap size of 0.5, 1.0, and 1.5 mm, and fix the Young's modulus at 20 MPa, wafer pressure at 4 psi and ring pressure at 5 psi. As the ring pressure is 1 psi higher than the wafer pressure, the ring intrudes further into the pad than the wafer. The pad between the wafer and ring experiences two competing factors: it is pushed into the gap by the applied pressure, but it is lifted by the further intruded ring. When the gap is large, as the case for a gap of 1.5 mm, the ring only lifts the part of the pad near it and most of the pad bends into the gap. As the gap becomes smaller, the pad is lifted more. When the gap is small enough, as the case for a gap of 0.5 mm, the pad is not touching the wafer edge and the pressure is decreasing as it comes closer to the edge. Note that between the cases of large gap and small gap, the mean gap does not offer a uniform pressure distribution, as the case for a gap of 1.0 mm. Thus, it is not possible to achieve a uniform polishing by adjusting the gap size alone.

Third, we focus on the influence of applied pressure. We simulate for the cases of wafer pressures of 2, 4, and 6 psi, and with the ring pressure 1 psi larger than the wafer pressure in each case. The effective Young's modulus is fixed at 20 MPa, and the gap size at 1.0 mm. Again we observe the competing factors of applied pressure and the gap size. As the applied wafer pressure decreases from 6 psi to 2 psi, the contribution from the ring becomes more significant. The pad between the gap rises from being bent down to not touching the wafer edge, and as a result, the pressure distribution evolves from peaks at the edge to zero edge pressure.

In summary, the simulations of the static cases show that the typical pressure distribution peaks at the wafer edge. There are two competing factors: the applied pressure pushes the pad into the gap while the ring lifts it. As a result, when gap size or applied pressures become smaller, the pressure at the edge decreases. For the flat surface, the pressure distribution is not sensitive to the value of effective Young's modulus.

Unlike the static cases which focus on the pressure distribution at one time instant, 5h3 dynamic cases study the time evolution of the surface profile during CMP. For all of the dynamic cases, we vary the value of one factor while keeping the other two constant. A sixty-second simulation is performed, and the graphs show the surface profiles every five seconds. The initial profile is always assumed to be flat, and the topography variation after the sixty-second CMP is marked on the graph for comparison.

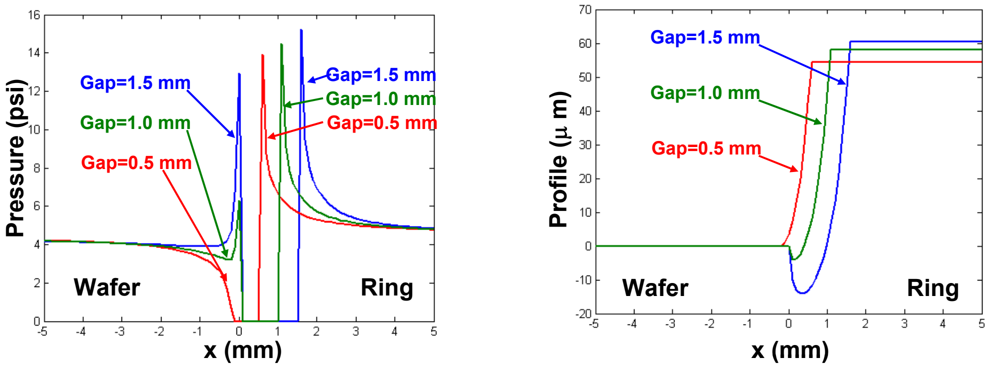

Figure 7. Static case to study the effect of gap size. For all three simulations, Young's modulus is 80 MPa, wafer pressure is 4 psi, and ring pressure is 5 psi.

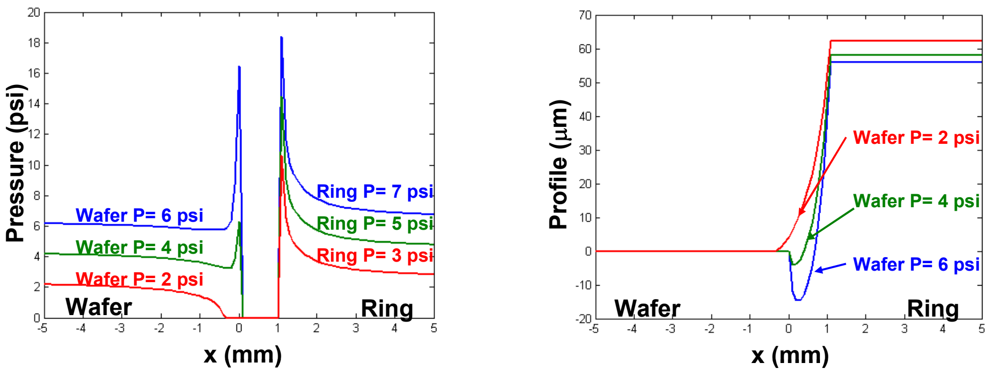

Figure 8. Static case to study the effect of pressures. For all three simulations, gap is 1 mm, Young's modulus is 80 MPa.

First, we simulate for different values of effective Young's modulus, 80 MPa and 200 MPa, and the wafer pressure is kept at 6 psi, ring pressure at 7 psi and gap size at 1.5 mm. From figure 9, we observe that both cases have a similar profile shape, but the stiffer pad (200 MPa) results in smaller roll-off value, 0.37 μm compared to 0.62 μm for the softer pad. Although Young's modulus is not a sensitive factor in the analysis of static cases, a stiffer pad requires less topography change to offset the non-uniform pressure distribution. Thus in the dynamic case, larger Young's modulus pads result in less edge roll-off as expected.

Next we study how the surface evolution changes in response to different gap sizes; the simulation results are shown in figure 10. In the simulations, we choose gap sizes of 1.5 mm and 0.5 mm, while the wafer pressure is kept at 6 psi, ring pressure at 7 psi, and effective Young's modulus at 80 MPa. In the static simulation, a large gap results in pressure peaked at the edge. Thus, we expect excessive polishing at the edge resulting in edge roll-off profiles. For smaller gaps, we use different pressure values in the simulation from the ones in static cases. The dynamic case seems to be better matched with the median gap size in the static case, in which the pressure distribution has a small peak at the edge and a minimum value about 0.3 mm inside. The profile evolution in the dynamic case with smaller gap size shows a similar pattern, which is slightly over-polishing at the wafer edge, and the slowest polishing occurs about 0.3 mm inside the edge.

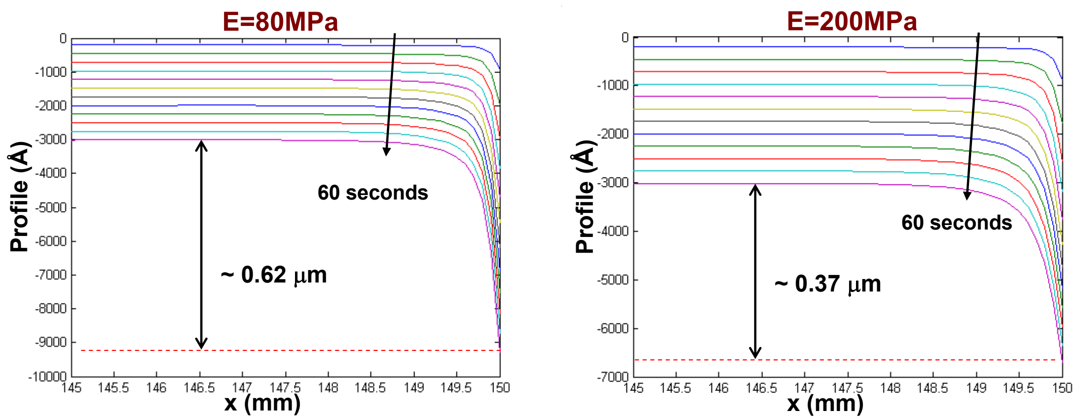

Figure 9. Dynamic case to study the effect of Young's modulus. For both simulations, gap is 1.5 mm, wafer pressure is 6 psi, and ring pressure is 7 psi.

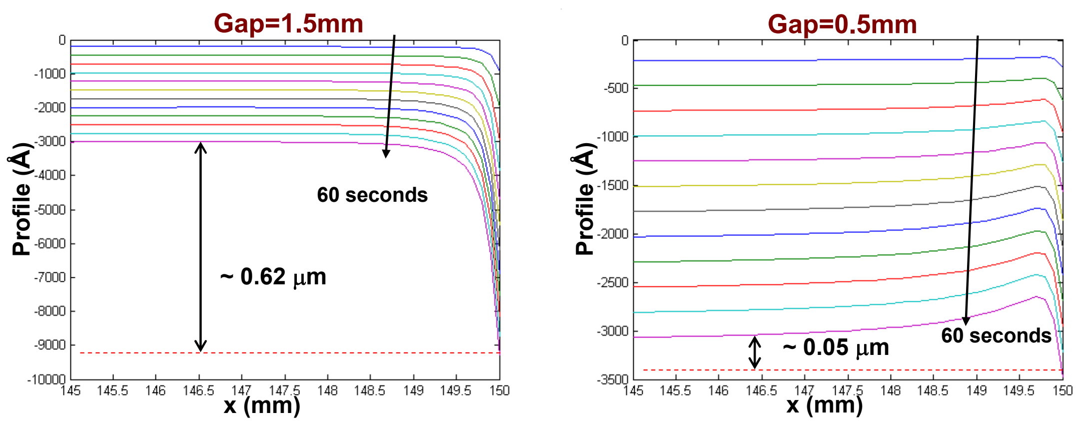

Figure 10. Dynamic case to study the effect of gap size. For both simulations, Young's modulus is 80 MPa, wafer pressure is 6 psi, and ring pressure is 7 psi.

Third, the simulations of different pressures are done by fixing the gap size at 1.5 mm and the effective Young's modulus at 80 MPa. The ring pressure is always kept 1 psi higher than the wafer pressure, and the two c use wafer pressure of 6 psi and 2 psi. The larger pressure results in excessive edge over-polishing, which is consistent with the static simulation result. The smaller pressure results in slight over-polishing at the edge and the slowest polishing taking place about 0.5 mm inside, which seems to match the median pressure case in the static cases.

These dynamic simulations suggest that: the larger the gap size, the faster the edge polishes; the higher the pressure, the faster the edge polishes; and the stiffer the pad, the more uniform the wafer polishes. A combination of large gap size, large applied pressure, and small Young's modulus will result in fast edge-polishing evolution, which is the case in figure 9 (left). A combination of small gap size, small applied pressure and small Young's modulus will result in a slow edge polishing. We choose gap size 0.5 mm, wafer pressure 2 psi, ring pressure 3 psi, and Young's modulus 80 MPa, and the result is shown in figure 12 (left). A uniform polishing can be achieved by a combination of median gap size, median pressures, and large Young's modulus. Figure 12 (right) shows the simulation result for gap size 1.0 mm, wafer pressure 4 psi, ring pressure 5 psi, and Young's modulus 200 MPa.

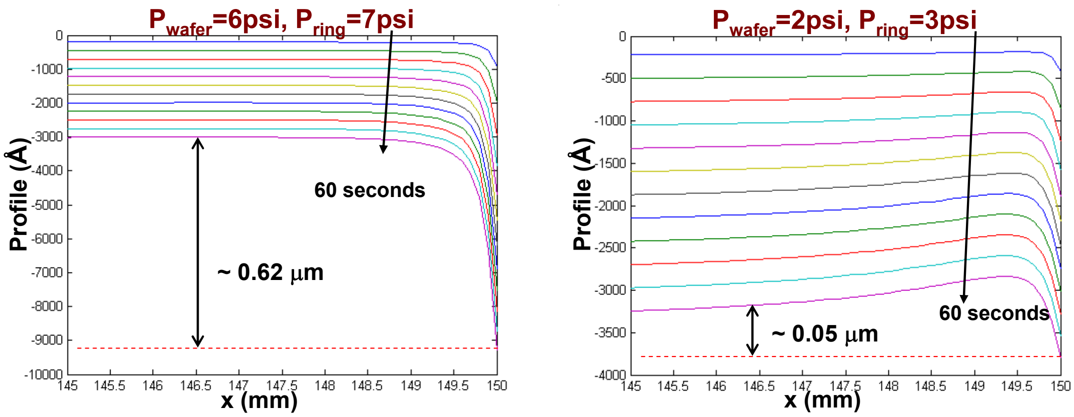

Figure 11. Dynamic case to study the effect of pressures. For both simulations, gap is 1.5 mm, and Young's modulus is 80 MPa.

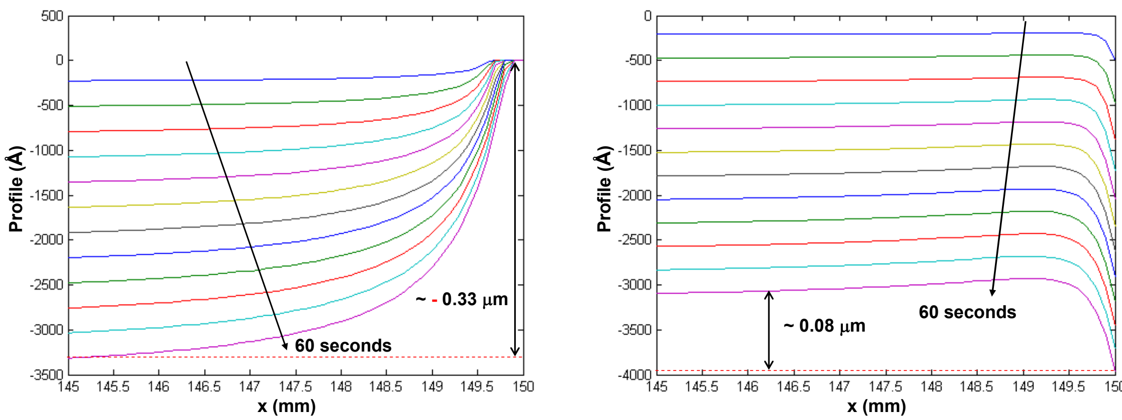

Figure 12. Left plot shows simulated surface evolution using a slow edge polishing setup: Young's modulus is 80 MPa, gap size is 0.5 mm, wafer pressure is 2 psi, and ring pressure is 3 psi. Right plot shows the surface evolution using a more uniform polishing setup: Young's modulus is 200 MPa, gap size is 1.0 mm, wafer pressure is 4 psi, and ring pressure is 5 psi.

In figure 13, the measured thickness profiles near the wafer edge are compared with simulated profiles after 60-second polishing. The simulated results capture the range of observed variations in measured edge roll-off profiles.

We have studied how the edge roll-off profiles can be generated from the CMP process; the next question is how the initial topography profile will affect the performance of later CMP steps. For a flat initial surface, as we studied earlier, a combination of median gap size, median pressures, and stiffer pad will give a nearly uniform polishing.

To study the impact on CMP of a wafer with a starting edge roll-off profile, we will consider two sets of CMP parameters. Both cases have the same effective Young's modulus, but the first set has higher pressures and larger gap. As discussed earlier, the first set should result in faster edge polishing, and the simulation results do indeed show a similar result as seen in figure 14. The first case shows the surface evolving from a starting roll-off to a final profile with a similar edge roll-off pattern. The second case polishes slower on the edge, and the surface becomes flatter during CMP, as seen at right in figure 14. Although the second case reduces the surface non-uniformity, its removal amount is more non-uniform than that of the first one (figure 15).

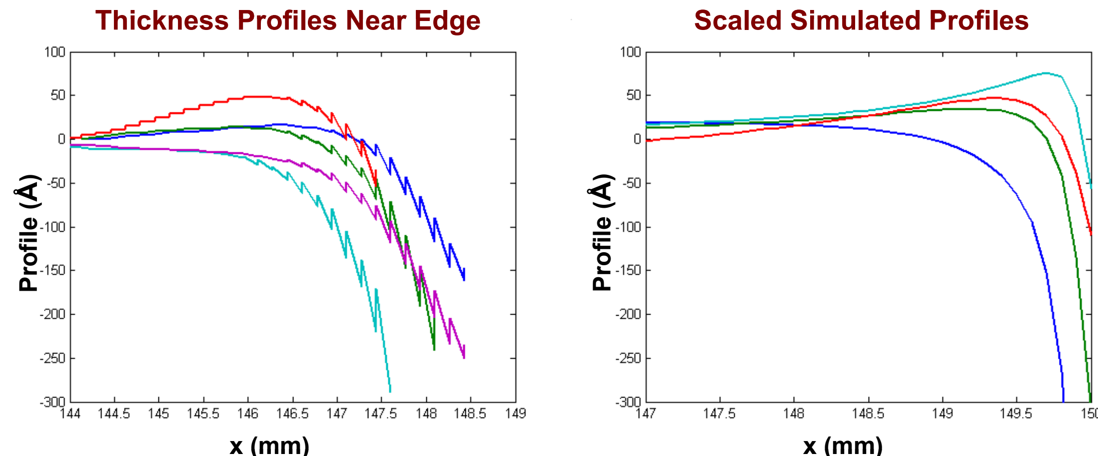

Figure 13. Left plot shows the measured thickness profiles near the wafer edge, and the right plot shows the simulated profile after 60-second CMP. The simulated profiles are able to capture different types of observed behaviors in measured edge profile. Note: the measuremed profile shows zig-zag artifacts due to resolution limits that are not present in the actual surface.

Thus, if the objective is to improve the surface flatness, (e.g., as preparation for a lithography process), it is better to match a starting edge roll-off profile with a slow edge polishing setup. If the objective is uniform polishing of a surface film (for example), then it is better to match the starting edge roll-off profile with a fast edge polishing setup. Similarly, for an initial edge "roll-up" profile, it is better to use a slow edge polishing setup to achieve more uniform polishing, as shown in figures 16 and 17.

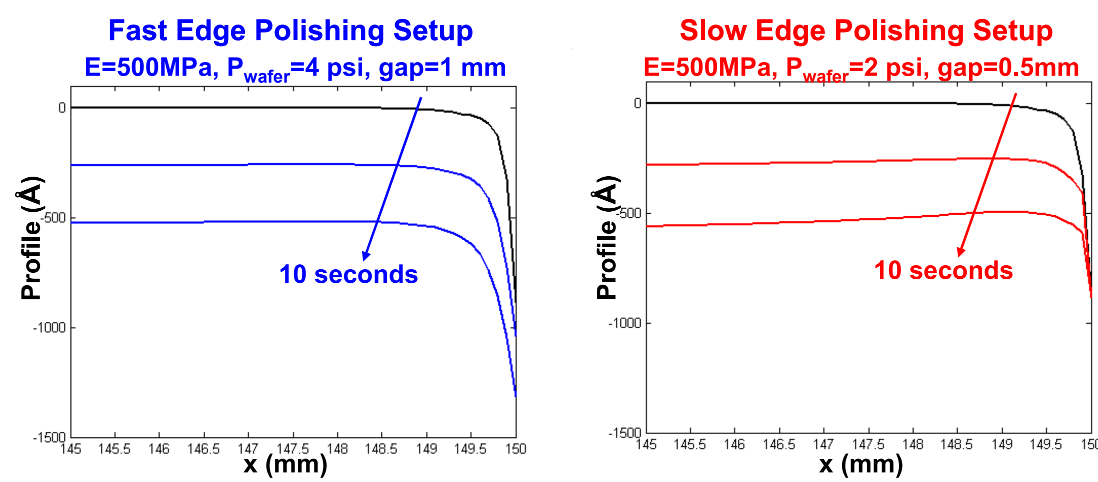

Figure 14. Dynamic simulations starting from initial edge roll-off profiles. The left plot shows simulation using a fast edge polishing setup, and the surface shape does not change significantly. The right plot shows simulation with a slow edge polishing setup, and the topography non-uniformity is reduced.

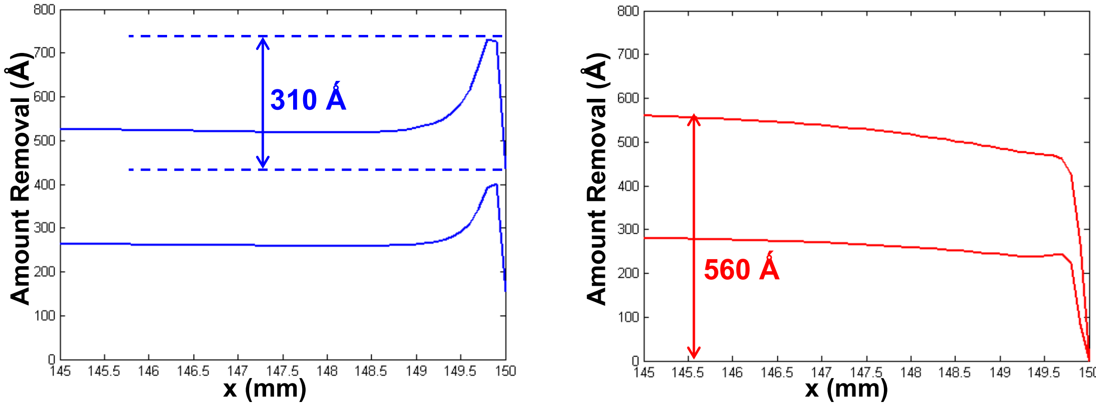

Figure 15. Same dynamic simulations as performed in figure 14, but the amount removed is shown.

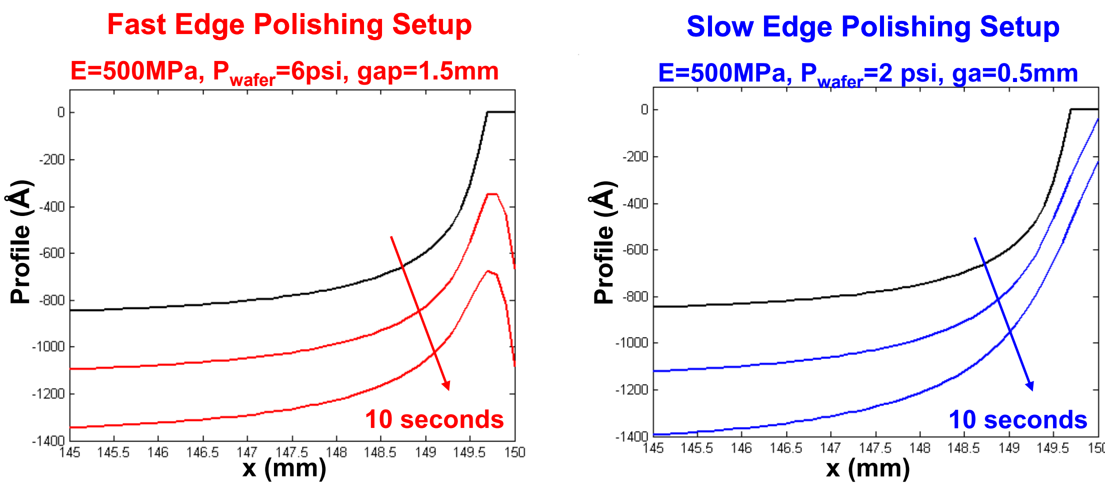

Figure 16. Dynamic simulations starting from an initial edge "roll-up" profile. The left plot shows simulation using a fast edge polishing setup, and the topography non-uniformity is thus reduced. The right plot shows simulation with a slow edge polishing setup, and the surface shape does not change significantly.

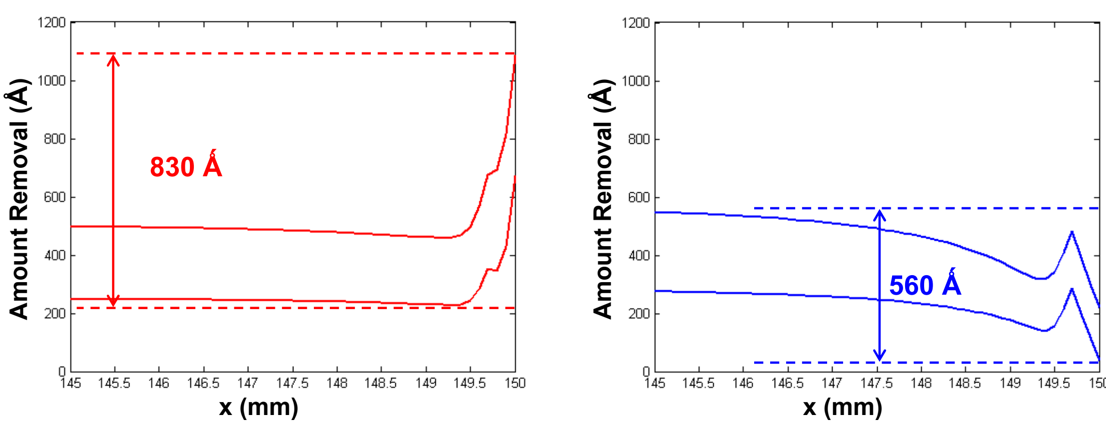

Figure 17. Same dynamic simulations as performed in figure 16, but the amount removed is shown.

CONCLUSION

In this paper, we study how a wafer edge roll-off profile can be generated during CMP and how existing edge roll-off affects further CMP process steps. A contact wear model is applied to simulate the CMP process near the wafer edge in an approximate fashion. The three factors studied are the effective Young's modulus of polishing pad, gap size between wafer and retaining ring, and the pressures on the wafer and the ring. The simulation results suggest better edge uniformity can be achieved by using stiffness tailored pads, median gap size, and median pressures. Edge polishing is also influenced by the initial surface profile. Better matching of the starting profile and tool set, such as edge roll-off profile with fast or slow edge polishing tool setups, will provide more uniform post-polish results. In future work, obtaining experimental data on pre- and post- CMP can help to test and calibrate the model predictions.

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